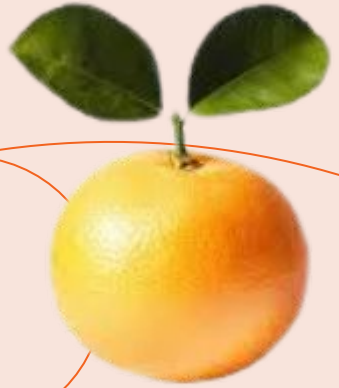
Search Algorithms

Vice President: Prisha Khanna

Agenda

https://www.codechef.com/ide



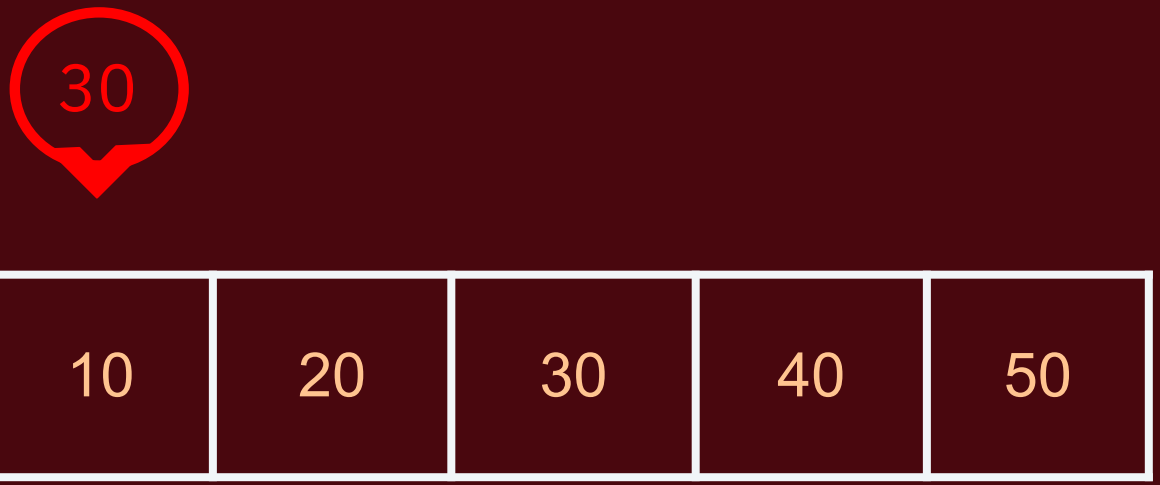
1. Linear Search
2. Binary Search
3. Interpolation Search
4. Jump Search Code

Linear Search

Linear Search

- Objective: Finds target element
- Checks each element till target element is found

Array = {10, 20, 30, 40, 50}
Key = 30



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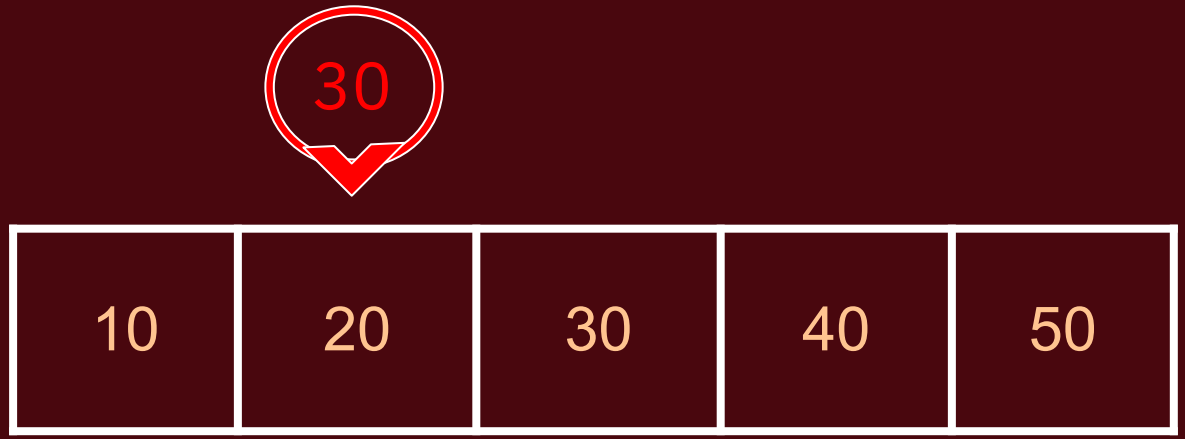

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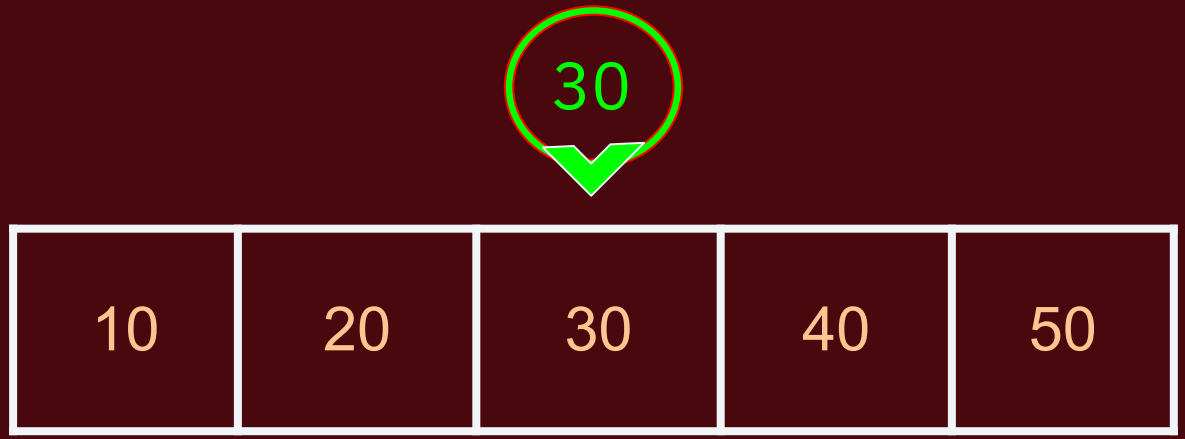


Linear Search Code

```
list = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

def linear_search(list, target):
    for i in range(0, len(list)):
        if list[i] == target:
            return i

print("Linear Search:")
print("The index of the integer 5 is " + str(linear_search(list, 5)))
```

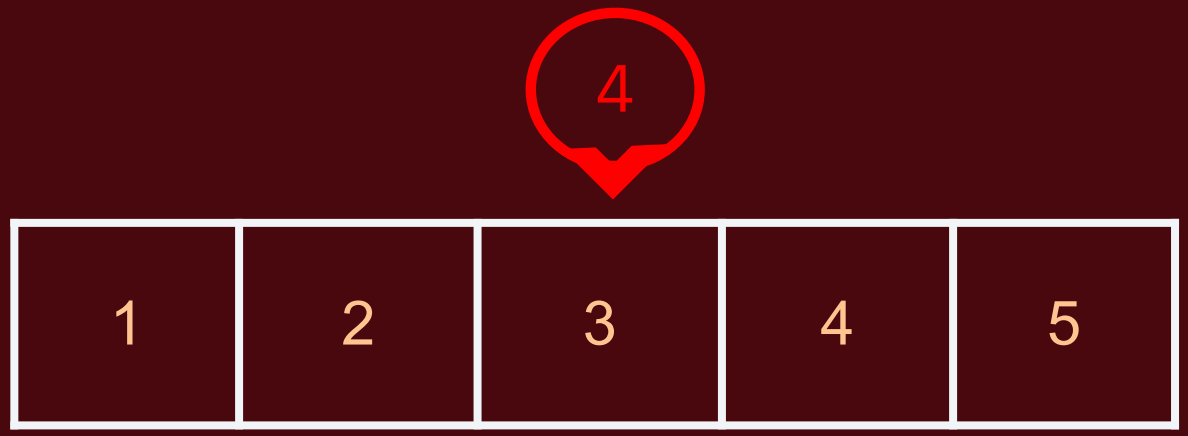
```
Linear Search:
The index of the integer 5 is 4
```

Binary Search

Binary Search

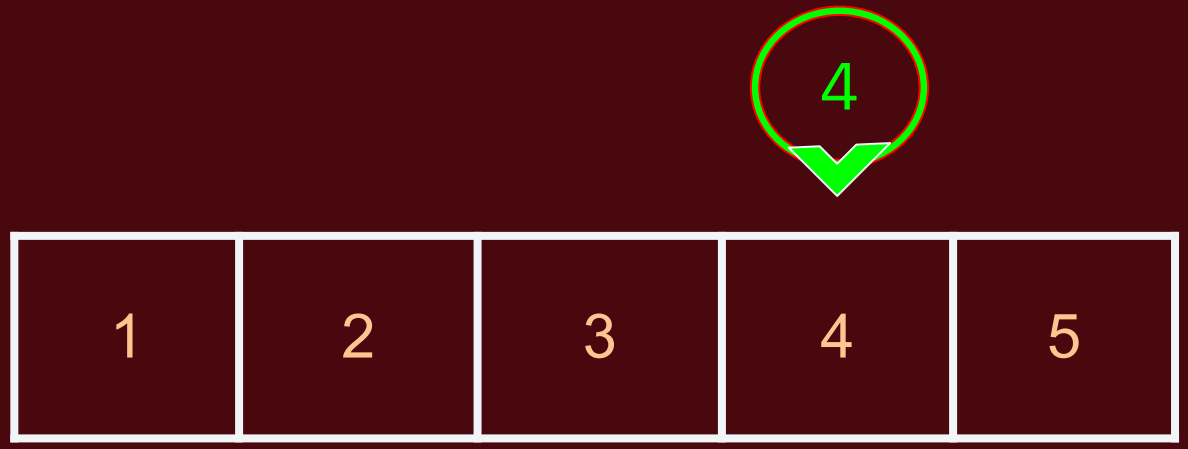
- Objective: Finds target element
- Starts in the middle and divides in halves
- If the Key is *bigger* than the middle it moves left
- If the Key is *smaller* than the middle it moves right

Array = {1, 2, 3, 4, 5}
Key = 4



4 is greater than 3 so the search space moves to the right

Array = {1, 2, 3, 4, 5}
Key = 4



It moves to the middle of the next segment and finds 4

Binary Search

```
def binary_search(list, target):
    left_bound = 0
    right_bound = len(list) - 1
    while left_bound <= right_bound:
        mid_point = int((left_bound + right_bound)/2)
        if list[mid_point] == target:
            return mid_point
        elif list[mid_point] < target:
            left_bound = mid_point + 1
        else:
            right_bound = mid_point - 1

print("Binary Search:")
print("The index of the integer 4 is " + str(binary_search(list, 4)))
```

```
Binary Search:
The index of the integer 4 is 3
```

Interpolation Search

Interpolation Search

- Objective: Narrow down search space to find target
- Starts at the probable location, not the middle

Interpolation Search
33
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
1 3 7 10 14 15 16 18 20 21 22 23 25 33 35 42 45 47 50 52
33 >23 Set low = 12
Total Comparisons: 1

Interpolation

This code is contributed by Hardik Jain

* GeeksforGeeks

```
lef interpolationSearch(arr, lo, hi, x):

    # Since array is sorted, an element present
    # in array must be in range defined by corner
    if (lo <= hi and x >= arr[lo] and x <= arr[hi]):

        # Probing the position with keeping
        # uniform distribution in mind.
        pos = lo + ((hi - lo) // (arr[hi] - arr[lo]) *
                    (x - arr[lo]))

        # Condition of target found
        if arr[pos] == x:
            return pos

        # If x is larger, x is in right subarray
        if arr[pos] < x:
            return interpolationSearch(arr, pos + 1,
                                       hi, x)

        # If x is smaller, x is in left subarray
        if arr[pos] > x:
            return interpolationSearch(arr, lo,
                                       pos - 1, x)
    return -1

 Driver code


 Array of items in which
 search will be conducted
rr = [10, 12, 13, 16, 18, 19, 20,
      21, 22, 23, 24, 33, 35, 42, 47]
 = len(arr)

 Element to be searched
 = 18
ndex = interpolationSearch(arr, 0, n - 1, x)

f index != -1:
    print("Element found at index", index)
lse:
    print("Element not found")
```

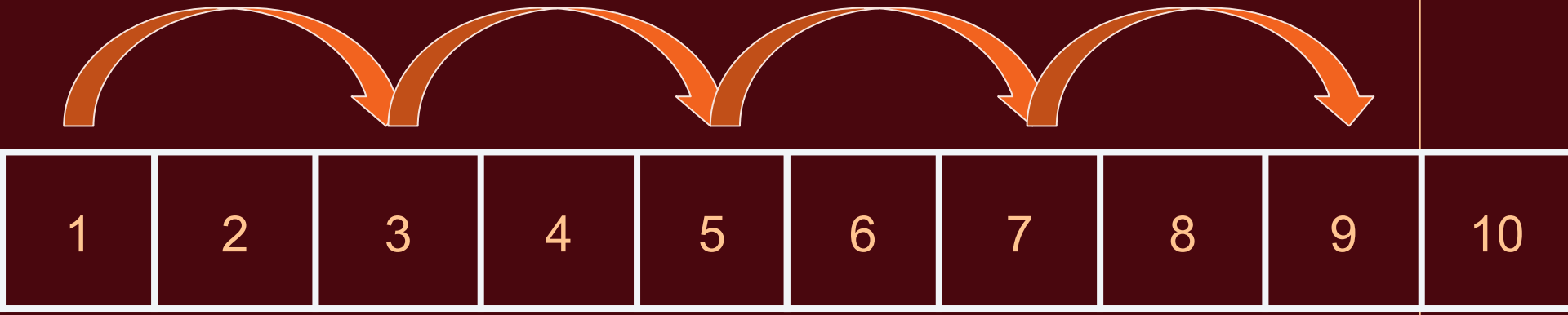
```
Element found at index 4
```

Jump Search

Jump Search

- Objective: Jump ahead till it finds target
- Starts at the first index
- Checks fewer elements by skipping some

Array = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
Key = 9



It moves to the middle of the next segment and finds 4

Jump Search

```
def jump_search(list, target):
    n = len(list)
    step = int(math.sqrt(n))
    prev = 0
    while list[min(step, n) - 1] < target:
        prev = step
        step += int(math.sqrt(n))
        if prev >= n:
            return -1
    while list[prev] < target:
        prev += 1
        if prev == min(step, n):
            return -1
        elif list[prev] == target:
            return prev

print("Jump Search:")
print("The index of the integer 6 is " + str(jump_search(list, 6)))
```

```
Jump Search:
The index of the integer 6 is 5
```

Thank you!

Please come again on Thursday!
Library IHS

Contact
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Sources

https://www.geeksforgeeks.org/dsa/linear-search/

https://www.geeksforgeeks.org/dsa/binary-search/

https://www.geeksforgeeks.org/dsa/interpolation-search/

https://www.geeksforgeeks.org/dsa/jump-search/